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futchhd.06@tamu.edu
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Math 151, section 510-518, Fall 2024

Math 151 Homework Section 3.2 (Homework)

INSTRUCTOR
Sang Lee
Texas A&M University

Current Score

QUESTION	1	2	3	4	5	6	7	8	9	10	11	12	13
POINTS	–/1.87	–/1.87	–/1.87	–/1.87	–/1.87	–/1.87	–/1.87	–/1.87	–/1.87	–/1.87	–/1.87	–/1.87	–/1.87

TOTAL SCORE

–/30	0.0%

Due Date

WED, SEP 25, 2024
11:55 PM CDT

Request Extension

Assignment Submission & Scoring

Assignment Submission

For this assignment, you submit answers by question parts. The number of submissions remaining for each question part only changes if you submit or change the answer.

Assignment Scoring

Your best submission for each question part is used for your score.

1. **[–/1.87 Points]** DETAILS MY NOTES **SCALCET9 3.2.001.**

Let $f(x) = (1 + 3x^2)(x - x^2)$.

Find the derivative by using the Product Rule.

$f'(x) =$

Find the derivative by multiplying first.

$f'(x) =$

Do your answers agree?

- Yes
- No

2. **[–/1.87 Points]** DETAILS MY NOTES **SCALCET9 3.2.003.**

Differentiate.

$$y = (3x^2 + 5)(7x + 4)$$

$y' =$

3. **[–/1.87 Points]** DETAILS MY NOTES **SCALCET9 3.2.008.**

Differentiate.

$$g(x) = \left(x + 5\sqrt{x}\right)e^x$$

$g'(x) =$

4. **[–/1.87 Points]** DETAILS MY NOTES **SCALCET9 3.XP.2.021.**

Differentiate.

$$R(t) = (t + e^t)(5 - \sqrt{t})$$

$R'(t) =$

5. **[–/1.87 Points]** DETAILS MY NOTES **SCALCET9 3.XP.2.004.**

Differentiate.

$$y = \frac{x^5}{4 - x^4}$$

$y' =$

6. **[–/1.87 Points]** DETAILS MY NOTES **SCALCET9 3.XP.2.017.**

Differentiate the function.

$$g(x) = \frac{9 + 4x}{1 - 6x}$$

$g'(x) =$

7. **[–/1.87 Points]** DETAILS MY NOTES **SCALCET9 3.2.009.**

Differentiate.

$$y = \frac{8x}{e^x}$$

$y' =$

8. **[–/1.87 Points]** DETAILS MY NOTES **SCALCET9 3.2.013.**

Differentiate.

$$f(t) = \frac{7t}{t^3 - t - 1}$$

$f'(t) =$

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9. **[–/1.87 Points]** DETAILS MY NOTES **SCALCET9 3.XP.2.024.**

Find $f'(x)$ and $f''(x)$.

$$f(x) = x^5e^x$$

$f'(x) =$

$f''(x) =$

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10. **[–/1.87 Points]** DETAILS MY NOTES **SCALCET9 3.XP.2.030.**

Find an equation of the tangent line to the given curve at the specified point.

$$y = \frac{2x}{x+2}, \quad (2, 1)$$

$y =$

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11. **[–/1.87 Points]** DETAILS MY NOTES **SCALCET9 3.2.047.MI.**

Let $f(x) = e^x g(x)$, where $g(0) = 3$ and $g'(0) = 4$. Find $f'(0)$.

12. **[–/1.87 Points]** DETAILS MY NOTES **SCALCET9 3.2.048.**

If $h(2) = 3$ and $h'(2) = -9$, find

$$\left.\frac{d}{dx}\left(\frac{h(x)}{x}\right)\right|_{x=2}.$$

13. **[–/1.87 Points]** DETAILS MY NOTES **SCALCET9 3.2.046.**

Suppose that $f(4) = 5$, $g(4) = 3$, $f'(4) = -4$, and $g'(4) = 2$. Find $h'(4)$.

(a) $h(x) = 3f(x) + 4g(x)$

$h'(4) =$

(b) $h(x) = f(x)g(x)$

$h'(4) =$

(c) $h(x) = \dfrac{f(x)}{g(x)}$

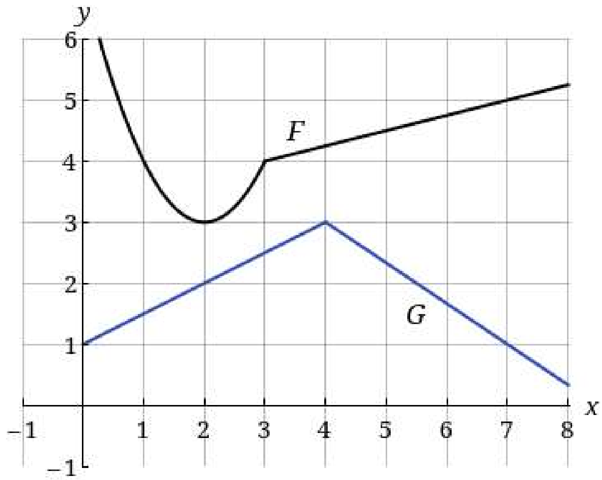
$h'(4) =$

(d) $h(x) = \dfrac{g(x)}{f(x) + g(x)}$

$h'(4) =$

14. **[–/1.87 Points]** DETAILS MY NOTES **SCALCET9 3.2.052.**

Let $P(x) = F(x)G(x)$ and $Q(x) = \dfrac{F(x)}{G(x)}$, where F and G are the functions whose graphs are shown.



(a) Find $P'(2)$.

(b) Find $Q'(7)$.

15. **[–/1.87 Points]** DETAILS MY NOTES **SCALCET9 3.2.027.**

Differentiate.

$$f(x) = \frac{x^9 e^x}{x^9 + e^x}$$

$f'(x) =$

16. **[–/1.95 Points]** DETAILS MY NOTES **SCALCET9 3.2.056.**

Find equations of the tangent lines to the curve

$$y = \frac{x - 1}{x + 1}$$

that are parallel to the line $x - 2y = 2$. (Enter your answers as a comma-separated list of equations.)

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